

BGR Project

Team: Golden Volt

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1. Introduction and Required Specs

In this report, we will discuss the design and simulation of a band gap reference circuit with the following specs.

Table 1: Required specs table

Parameter	Name	Specification			Units	Comments
		Min	Typ	Max		
Power Supply Voltage	VDD	1.2		3.3	V	Wide range
Temperature	Temp	-40		125	C°	
Output Voltage	V_{out}		0.8		V	
Line Regulation	$\Delta V_{out}/\Delta V_{DD}$			100	μV	
Output Noise (10Hz – 10KHz)	V_{nrms}			50	μV	
Power Supply Rejection at 100 KHz	PSR_{100KHz}	40			dB	
Power Supply Rejection at 10 MHz	PSR_{10MHz}	20			dB	
Area	A			0.2	mm ²	
Startup Time	$t_{startup}$		50		μsec	
Untrimmed Output Voltage Accuracy	ΔV_{out}			± 5	%	
Trimmed Output Voltage Accuracy	ΔV_{out}			± 1	%	

2. Design Procedure

- At first, choosing the devices, we used the thick 2V devices, and for the topology, we decided to use Banba architecture^[1], as it has better accuracy and PVT immunity compared to all other topologies we tried, but to increase its PSR, we managed to use cascode PMOS devices for the current mirrors.
- To control power consumption, we had to choose $R_{4,7}$ to achieve the required Power consumption, the greater $R_{4,7}$ value leads to a larger Area and less current consumption, as shown in Figure 3: BGR core and output branches.
- Then to achieve a ZTAT V_{REF} we have to choose $R_5 = \frac{PTAT \text{ voltage slope}}{CTAT \text{ voltage slope}} * R_{7,4}$ and after some tuning, we achieve poly current $I = \frac{V_{EB0}}{R_{4,7}} + \frac{V_T \ln m}{R_5}$ which we can use in generating a ZTAT $V_{REF} = V_{EB0} * \frac{R_8}{R_{4,7}} + V_T \ln m * \frac{R_8}{R_5}$ with good accuracy.
- To achieve the required V_{REF} of 800 mV we have to scale R_8 since $V_{REF} \propto R_8$ so scaling R_8 With trimming, we will achieve the required 800 mV at all Process Corners, Supply, and temperature (PVT) variations.
- Startup circuit is designed to avoid the Zero solution, we had to use only one stacked Transistor due Startup time problems come with using 2 stacked transistors in Startup circuit, but using one stacked only cause a large current consumption in startup circuit at high supply values and so we used large L device by vertical stacking to achieve large effective length and therefore reduce the current consumption in startup circuit.
- For the current mirror devices, we chose their $\frac{g_m}{I_d}$ to achieve the largest V_A at 1.2 V_{DD} to get a high PSR, and the Cascode devices M_{11} we use large $\frac{g_m}{I_D}$ (large W/L) and a large length to get large $g_m r_o$ and so achieve large R_{out} and higher DC PSR. They are not thick devices, but the bias circuit manages to keep it less than a third of the supply value.
- The Op-Amp we used is a self-biased 5T-OTA op amp to equalize V_x and V_y required for the main idea of Banba architecture. We chose PMOS input pair 5T-OTA topology for the Op-Amp to have low power consumption, maintain the stability easily and to couple the supply ripples on the output of the Op-Amp so that the PSR is a voltage divider between R_8 and R_{out} as the PMOS current mirror is off with respect to the supply ripples which are on both the source and the gate. We managed to use large sizes and large g_m for the

input pair to decrease their flicker noise and decrease the input-referred noise and therefore the output-referred noise.

- We used 2 capacitors, first at the output of the Op-Amp to V_{DD} For maintaining stability, achieving the required PM, the second capacitor is at the output node V_{REF} , that's used to achieve the required PSR at high frequency and decrease the integrated thermal noise $\frac{KT}{C}$, but also try to avoid making it too large so that the startup time achieves the required $50 \mu s$.

3. Design Schematics and Verilog-A models

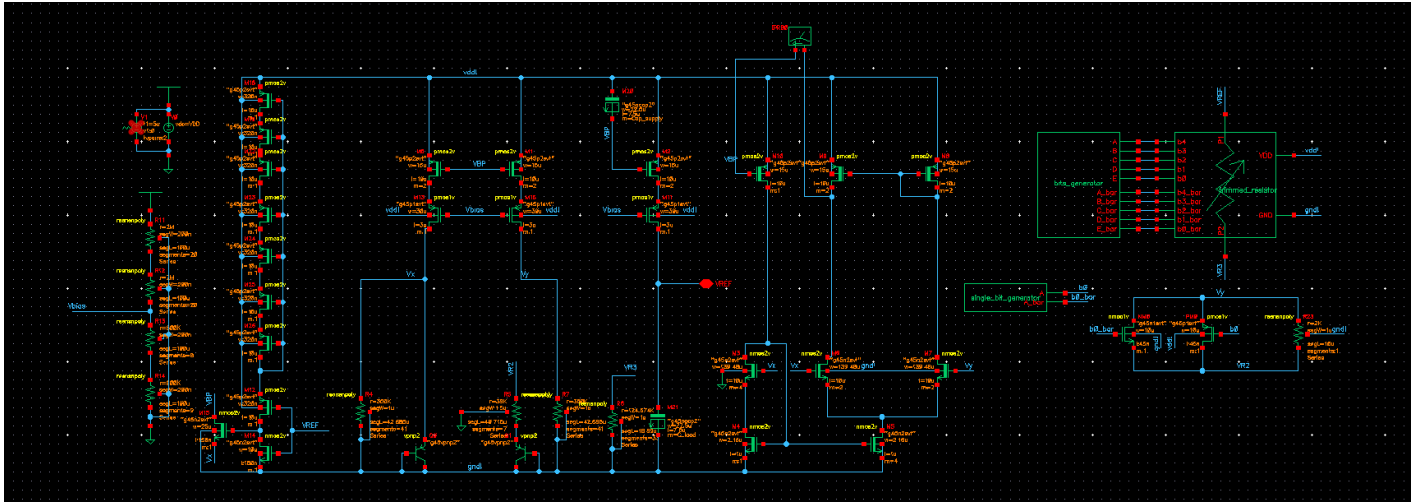


Figure 1: Full BGR schematic

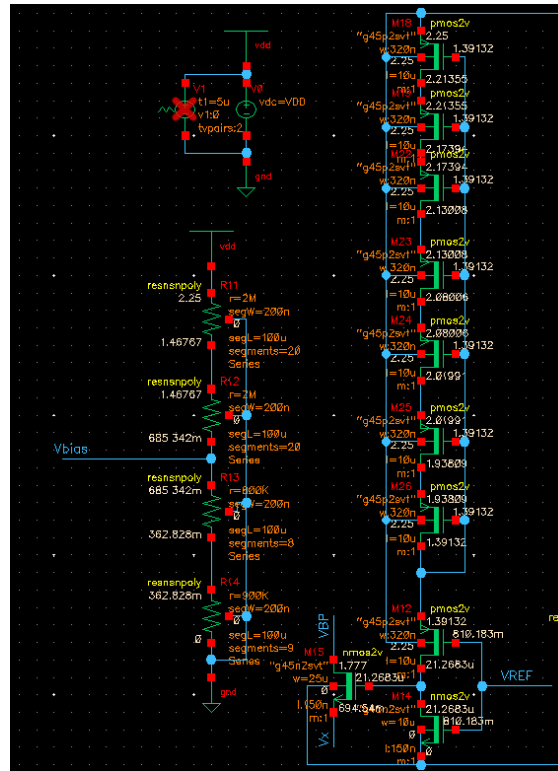


Figure 2: Bias and startup circuits

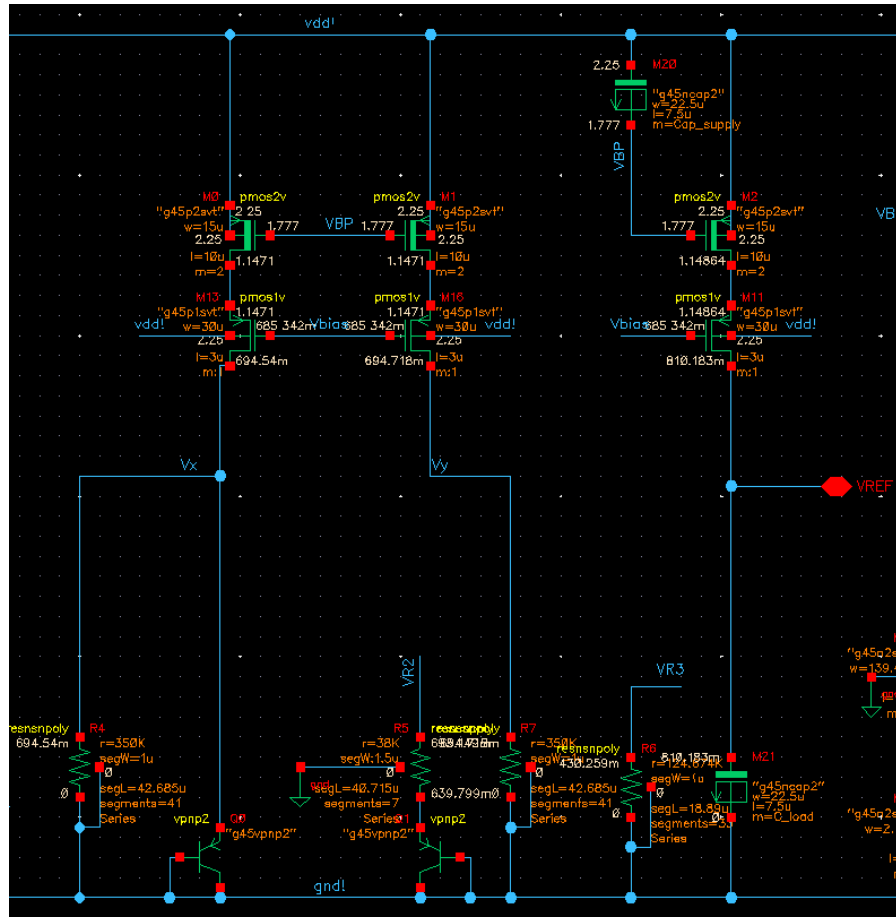


Figure 3: BGR core and output branches

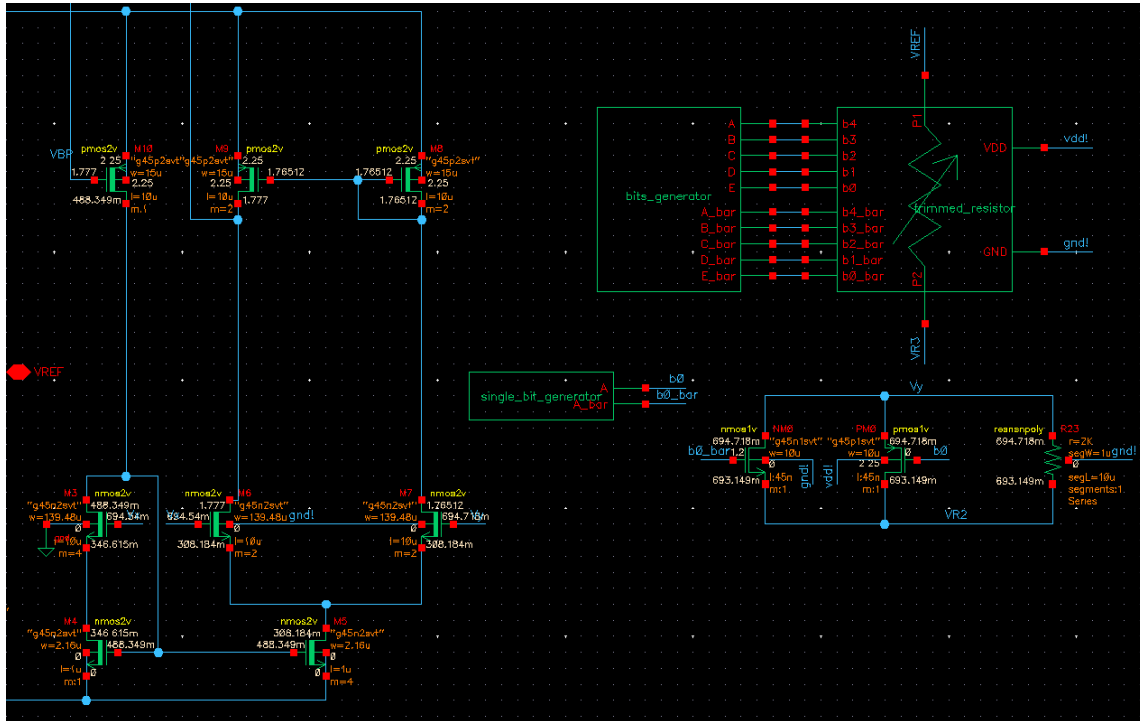


Figure 4: Self-biased OTA and trimming components

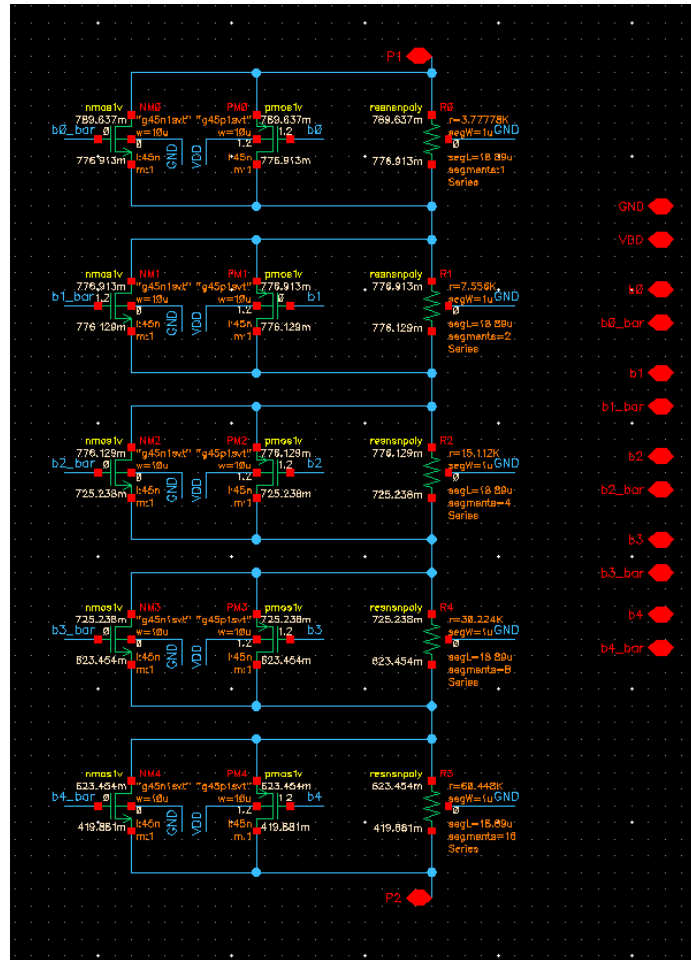


Figure 5: 5-bit trimming resistor schematic

```

`include "constants.vams"
`include "disciplines.vams"

module bits_generator (A, B, C, D, E, A_bar, B_bar, C_bar, D_bar, E_bar);

    // Explicit parameter (editable and overridable per-instance)
    parameter integer control_var = 0 from [0:31];

    // Outputs (electrical so you can connect to analog blocks)
    output A, B, C, D, E;
    output A_bar, B_bar, C_bar, D_bar, E_bar;
    electrical A, B, C, D, E, A_bar, B_bar, C_bar, D_bar, E_bar;

    // Output voltage levels & transition timing
    parameter real vlogic_high = 1.2;    // logic '1' voltage
    parameter real vlogic_low = 0.0;    // logic '0' voltage
    parameter real tdel = 0;
    parameter real trise = 0;
    parameter real tfall = 0;

    // integer temporaries for bit ops
    integer tmp;
    integer Ab, Bb, Cb, Db, Eb;

    analog begin

```

```

    tmp = control_var; // use integer tmp for bit ops

    // Extract bits: A = MSB, E = LSB
    Ab = (tmp >> 4) & 1;
    Bb = (tmp >> 3) & 1;
    Cb = (tmp >> 2) & 1;
    Db = (tmp >> 1) & 1;
    Eb = tmp          & 1;

    // Drive normal outputs
    V(A)      <+ transition(Ab ? vlogic_high : vlogic_low, tdel, trise,
tfall);
    V(B)      <+ transition(Bb ? vlogic_high : vlogic_low, tdel, trise,
tfall);
    V(C)      <+ transition(Cb ? vlogic_high : vlogic_low, tdel, trise,
tfall);
    V(D)      <+ transition(Db ? vlogic_high : vlogic_low, tdel, trise,
tfall);
    V(E)      <+ transition(Eb ? vlogic_high : vlogic_low, tdel, trise,
tfall);

    // Drive inverted outputs
    V(A_bar) <+ transition(Ab ? vlogic_low : vlogic_high, tdel, trise,
tfall);
    V(B_bar) <+ transition(Bb ? vlogic_low : vlogic_high, tdel, trise,
tfall);
    V(C_bar) <+ transition(Cb ? vlogic_low : vlogic_high, tdel, trise,
tfall);
    V(D_bar) <+ transition(Db ? vlogic_low : vlogic_high, tdel, trise,
tfall);
    V(E_bar) <+ transition(Eb ? vlogic_low : vlogic_high, tdel, trise,
tfall);
    end
endmodule

```

Figure 6: bits generator Verilog A model

```

`include "constants.vams"
`include "disciplines.vams"

module single_bit_generator (A, A_bar);

    // Explicit parameter (editable and overridable per-instance)
    parameter integer control_var = 0 from [0:1];

    // Outputs (electrical so you can connect to analog blocks)
    output A, A_bar;
    electrical A_bar;

    // Output voltage levels & transition timing
    parameter real vlogic_high = 1.2;    // logic '1' voltage
    parameter real vlogic_low  = 0.0;    // logic '0' voltage
    parameter real tdel        = 0;
    parameter real trise       = 0;
    parameter real tfall       = 0;

    // integer temporary for bit

```

```

integer Ab;

analog begin
    // Extract bit (just LSB of control_var)
    Ab = control_var & 1;

    // Drive normal output
    V(A)    <+ transition(Ab ? vlogic_high : vlogic_low, tdel, trise,
tfall);

    // Drive inverted output
    V(A_bar) <+ transition(Ab ? vlogic_low : vlogic_high, tdel, trise,
tfall);
end

endmodule

```

Figure 7: single-bit generator Verilog A model

4. Simulations and Results

4.1.DC OP Simulation

Total area equals 0.0374 mm^2 and typical current consumption (at 2.25 V supply) equals $18.07 \mu\text{A}$ so the typical power consumption is $40.66 \mu\text{W}$.

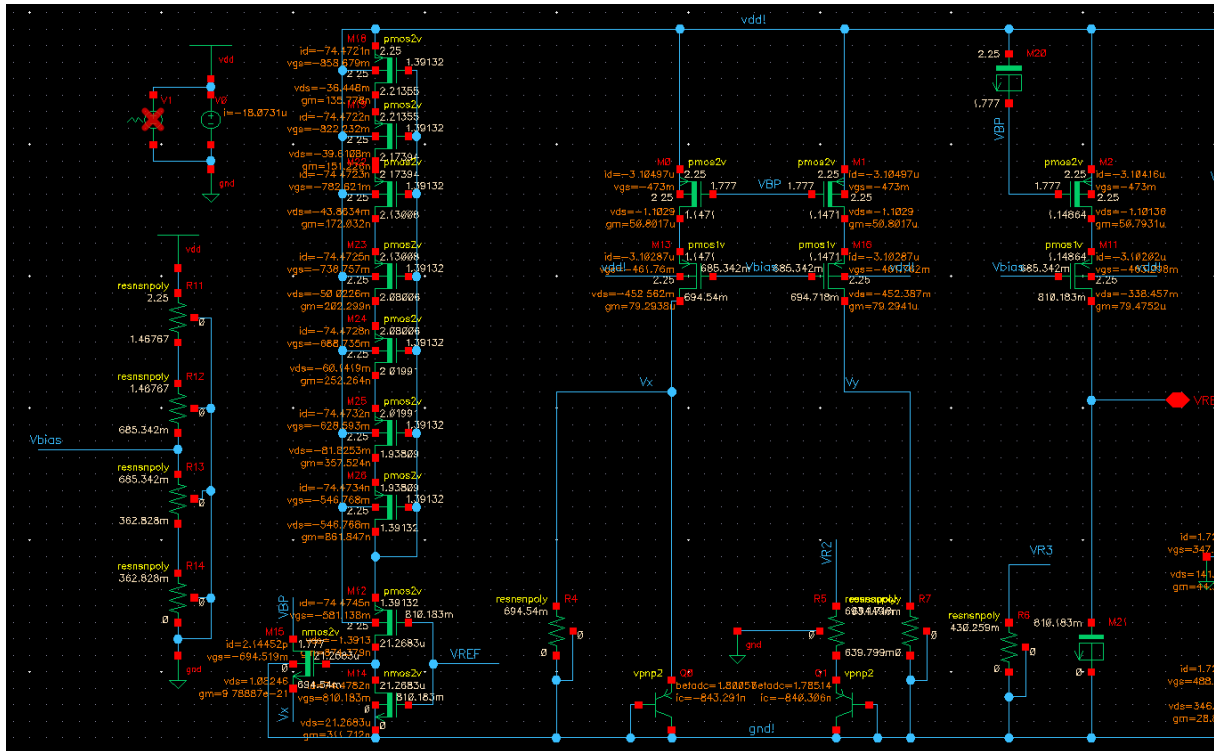


Figure 8: Left-half circuit DC OP

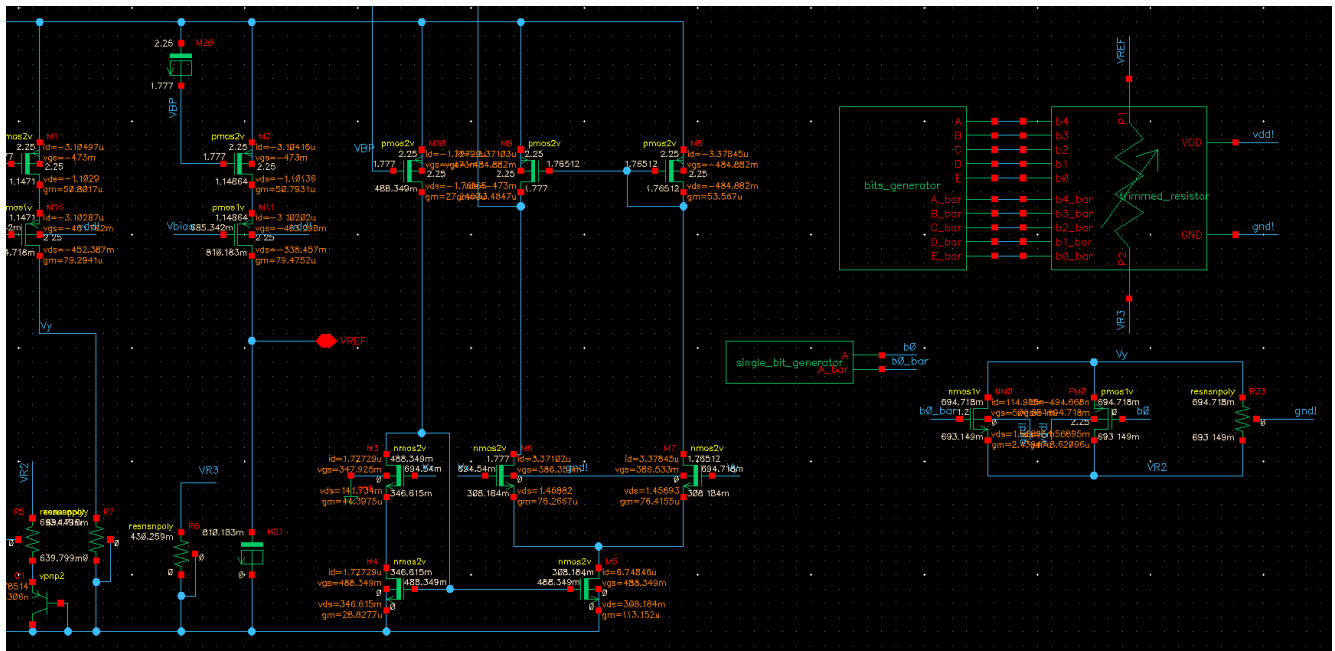


Figure 9: Right-half circuit DC OP

4.1. Before and after Trimming DC Simulations

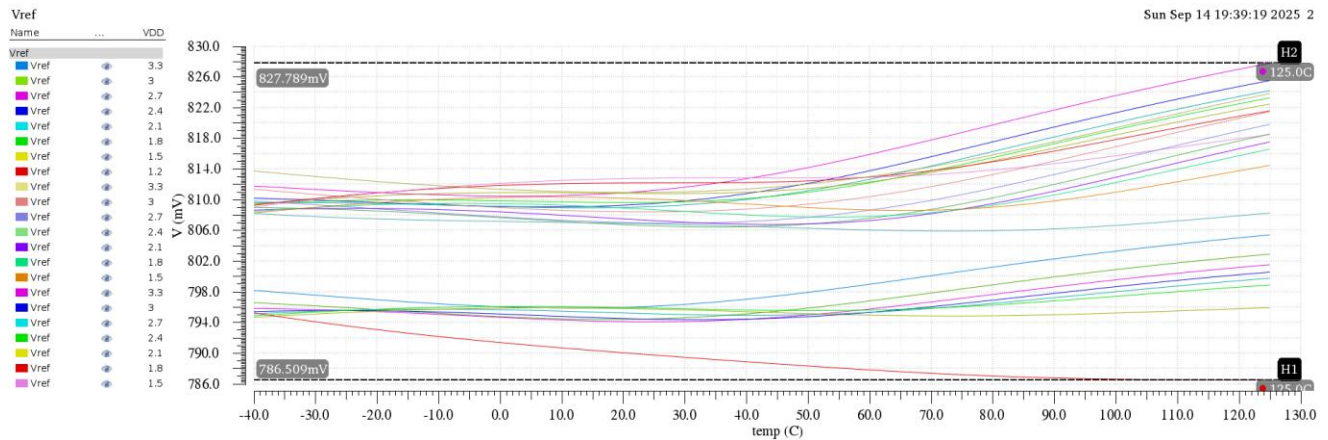


Figure 10: Untrimmed output variations vs temperature

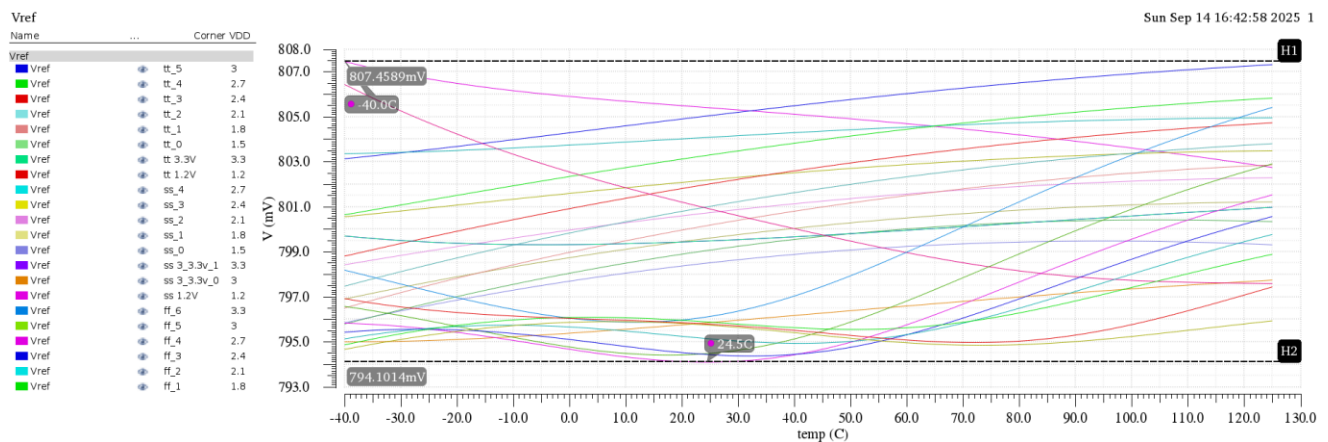


Figure 11: Trimmed output variations vs temperature

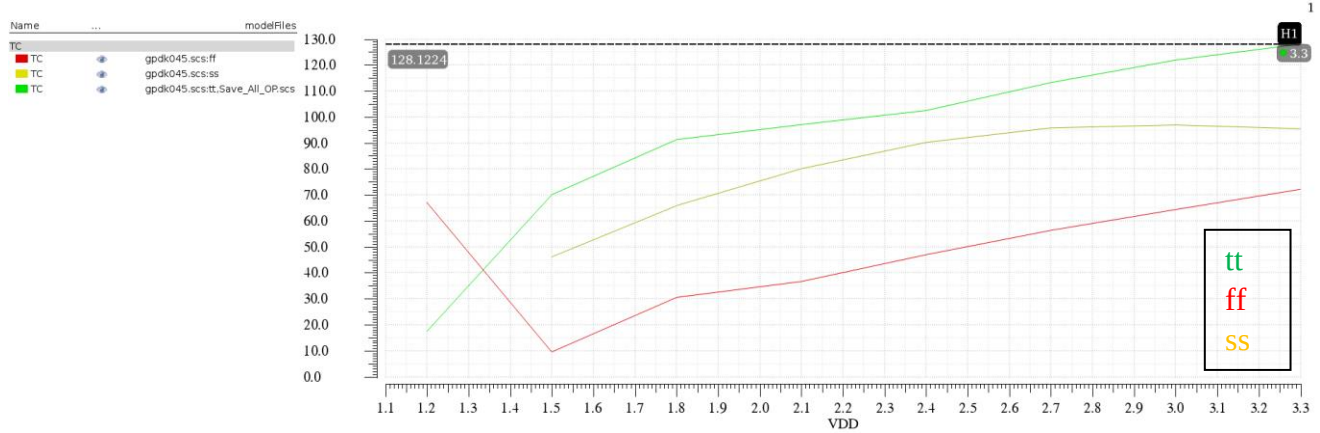


Figure 12: Untrimmed TC corners (in ppm/°C) vs VDD

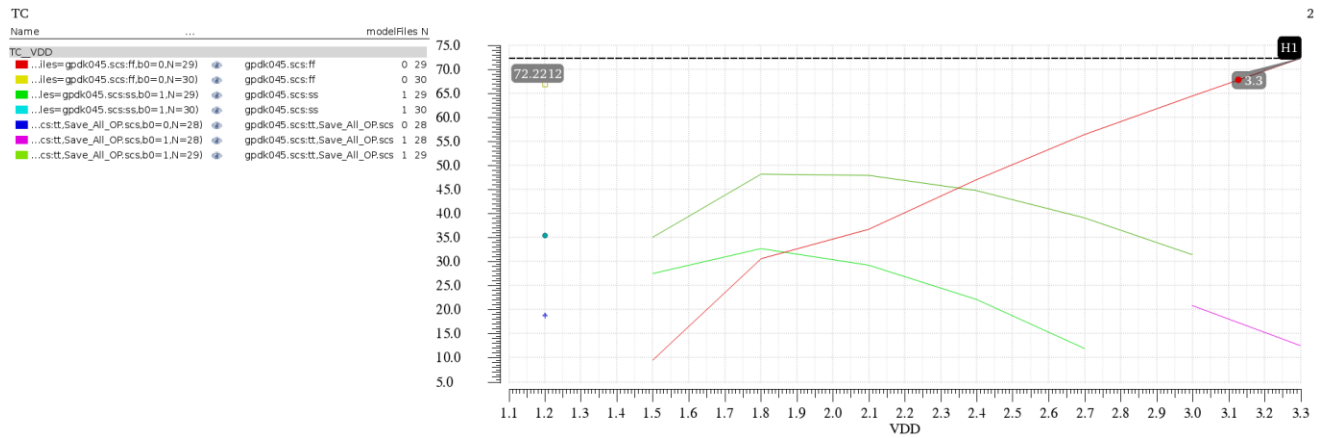


Figure 13: Trimmed TC corners (in ppm/°C) vs VDD

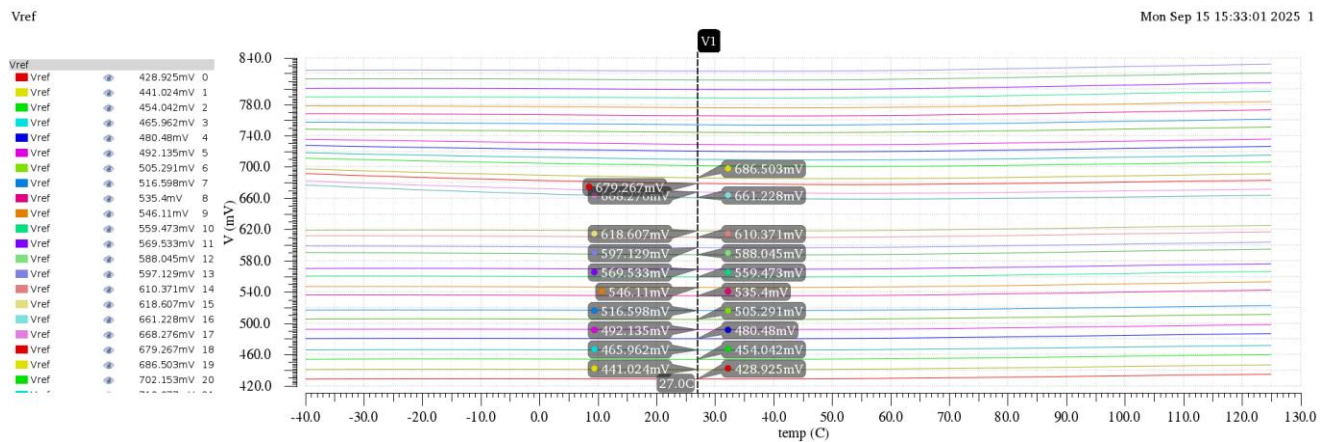


Figure 14: Output programmability vs temperature

4.4.Noise Simulation

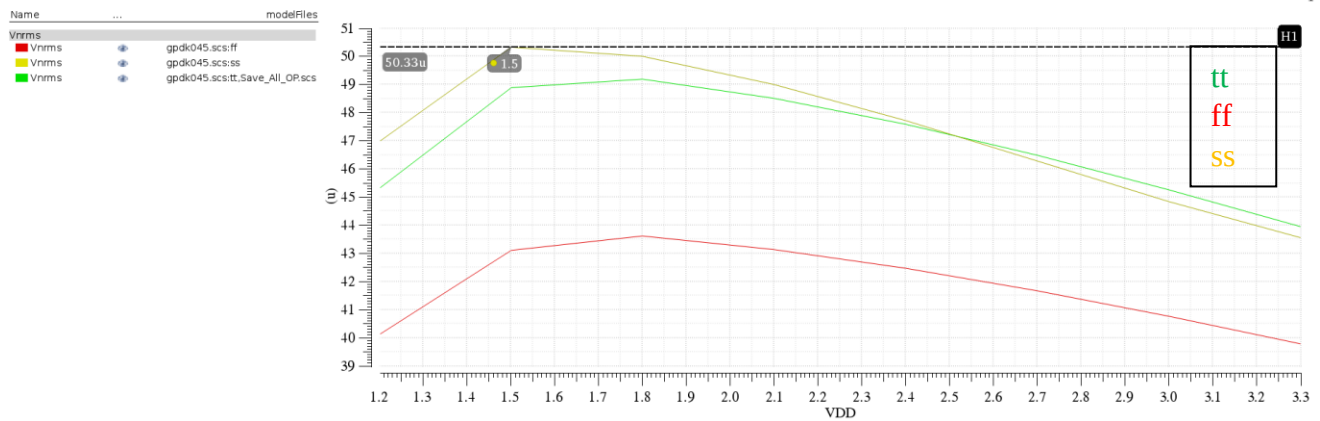


Figure 18: Vnrms corners (in V) vs VDD

4.5.Power Supply Rejection Simulation

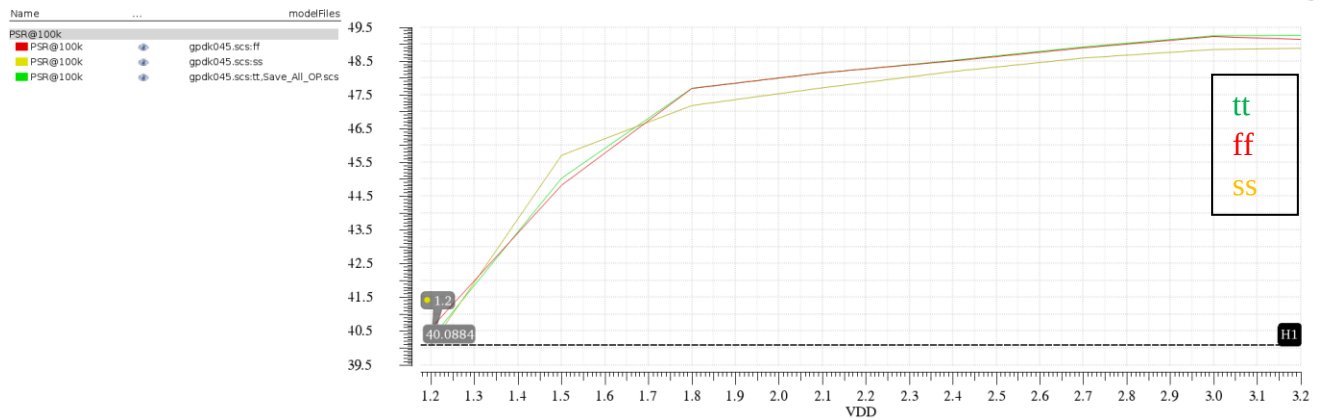


Figure 19: PSR @100 KHz corners (in dB) vs VDD

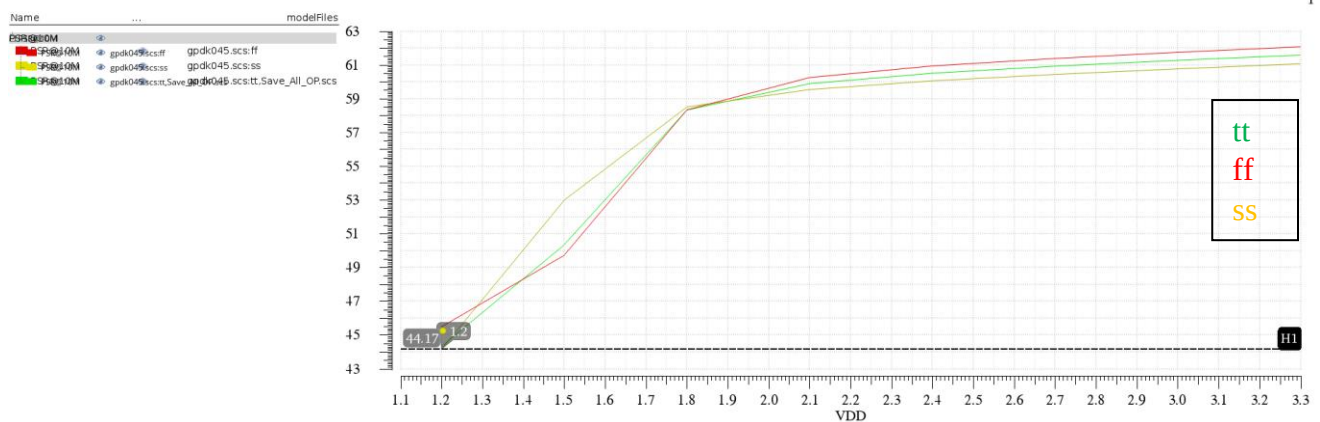


Figure 20: PSR @10 MHz corners (in dB) vs VDD

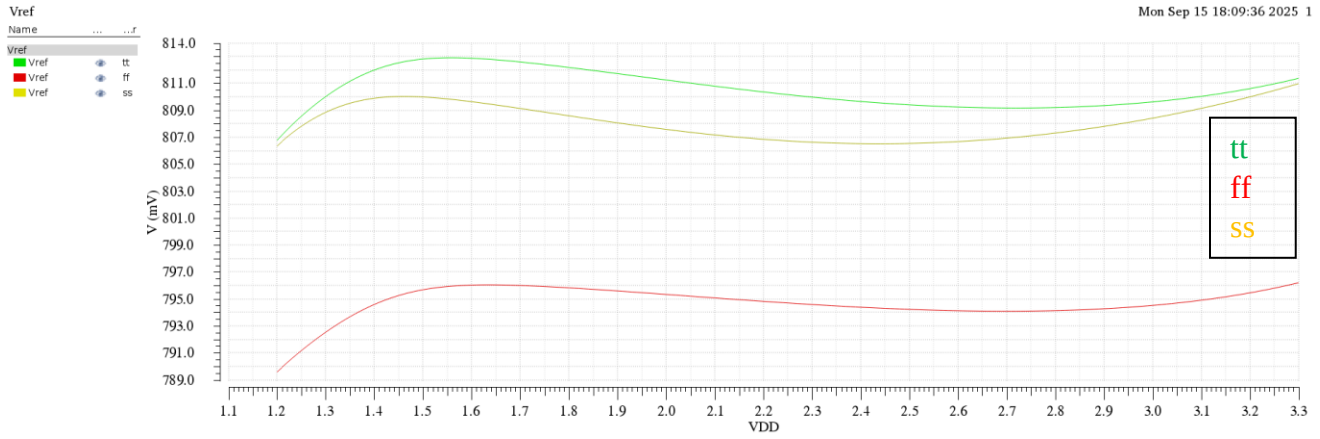


Figure 21: Line regulation corners

5. Final Results

Table 2: Final results table

Parameter	Name	Specification			Simulation Result			Units
		Min	Typ	Max	Min	Typ	Max	
Power Supply Voltage	VDD	1.2		3.3	1.2		3.3	V
Temperature	Temp	-40		125	-40		125	°C
Output Voltage	V_{out}		0.8		0.429	0.8	0.823	V
Power Consumption	PWR					40.66	59.63	μW
Line Regulation	$\Delta V_{out}/\Delta V_{DD}$			100			3132	μV
Output Noise (10Hz – 10KHz)	V_{nrms}			50			50.33	μV
Power Supply Rejection at 100 KHz	PSR_{100KHz}	40			40.09			dB
Power Supply Rejection at 10 MHz	PSR_{10MHz}	20			44.17			dB
Area	A			0.2			0.037	mm ²
Startup Time	$t_{startup}$		50			< 50		μsec
Untrimmed Output Voltage Accuracy	ΔV_{out}			± 5			± 3.5	%
Trimmed Output Voltage Accuracy	ΔV_{out}			± 1			± 0.94	%
Untrimmed Temperature Coefficient	TC						128.1	ppm/°C
Trimmed Temperature Coefficient	TC						72.22	ppm/°C
Figure of Merit	FOM					1.421		ppm.V/°C. μW

Note that $TC = \frac{V_{REF,max} - V_{REF,min}}{(V_{REF,max} + V_{REF,min})/2} * \frac{1}{\Delta Temp}$, $FOM = \frac{TC * V_{out}}{PWR}$.

6. Design challenges

The high range in VDD makes the design so hard to achieve the specs, and get good results in the whole range. Our main challenges were line regulation and output noise, where the line regulation required high R_{out} for the current mirror to get a high DC power supply rejection (PSR) for better line regulation, but the devices V_{DS} at 1.2 V supply was so small (0.4 V for both devices) So getting the required PSR in that case is so challenging, especially in the SS corner, as the PMOS current V_{SG} is higher, so the cascode device most probably enters the triode region, and the circuit becomes very sensitive to supply and temperature variations. We tried increasing the OTA gain but with no worthwhile change, tried device stacking to increase the effective length, and we got almost the same results. The PMOS cascode devices' bodies are connected to the supply, which gets better overall. R_{out} , where it suffers from body effect, causing the main mirror device to have less V_{DS} but it adds g_{mb} for the R_{out} expression which is 20 – 40% of g_m So the overall R_{out} It is better than the source-body connection and, therefore, the PSR. The ss corner suffers from the increase of V_{GS} so the mirror devices become more probable to go out of saturation, and that's a problem we handled by increasing the cascode devices' width to decrease their V_{GS} and maintain the desired operation at the ss corner.

Regarding the output noise, the spec is so challenging when the power consumption is limited to $20 \mu W$, where the mirror devices dominate the noise, and we tried to compromise between their flicker and thermal noise by changing their sizing and g_m but failed to achieve the spec and barely got $70 \mu V$ output noise, the OTA flicker noise was not a problem, so chopping is not effective, and we tried resistive degeneration to decrease the mirror noise, and it worked, but at large resistor values, which made the triode nearer to occur, even at large W/L, it did not maintain low PVT sensitivity. Finally, increasing the power consumption made the noise spec achievable, as we got higher g_m at the same biasing point for the circuit devices.

7. References

1. (H. Banba, H. Shiga, A. Umezawa, T. Miyaba, T. Tanzawa, S. Atsumi, and K. Sakui, "A CMOS bandgap reference circuit with sub-1-V operation," IEEE J. Solid-State Circuits, vol. 34, no. 5, pp. 670–674, May 1999.).